ServiceNow Certified System Administrator study guide

Organized by the CSA exam blueprint, with the distinctions the exam actually tests: which module, which table, which role, and which tempting answer is wrong.



Platform overview & navigation	**7%**	Instance configuration	**11%**
Applications for collaboration	**20%**	Self-service & automation	**20%**
Database management	**27%**	Data migration & integration	**15%**

60 questions in 90 minutes. Database Management alone is more than a quarter of the exam. Expect "choose two" and "choose three" questions; select exactly that many. Confirm the current blueprint on ServiceNow University before you book.

How to use this guide

This guide is a companion to the ServiceNow Administration Fundamentals e-book, not a replacement for it. Everything on the exam is covered in that book. Use this guide to check your understanding after each module and to drill the details the exam likes to test.

> **Watch for:** boxes like this mark distractors and traps that show up again and again.

> **Key point:** boxes like this mark facts you should be able to recall cold.

Do everything you read about in a Personal Developer Instance (PDI). Clicking through a module once is worth more than reading about it three times.

Client vs. server

Knowing where something runs eliminates wrong answers fast.

Client side (the browser)

Controls forms and fields as the user works

- Client Scripts (onLoad, onChange, onSubmit, onCellEdit)

Server side (the instance)

Controls data no matter how it arrives

- Business Rules
- Data Policies

- UI Policies
- Catalog Client Scripts and Catalog UI Policies
- Access Control (ACL) scripts
- Flows, Script Includes, transform maps

A **UI Policy** only affects data entered on a form. A **Data Policy** applies to forms, import sets, and web services alike.

1. Platform overview and navigation 7%

ServiceNow is a cloud platform. Scripting uses **JavaScript**. Response time for a user is the **round trip** between client and server: a **request** plus a **response**.

Interfaces and the banner

The platform includes three main interfaces: the **Now Platform user interface** (UI16 / Next Experience), **Service Portal**, and the **Now Mobile** apps.

Banner item	What it does
Instance logo	Returns you to your home page
Global search	Searches across the instance
Connect (chat bubbles)	Opens Connect chat. The full-screen view is **Connect > Connect Workspace**.
Help (question mark)	Help panel: guided tours, docs, and help on inputs and outputs in a flow
Gear (settings)	User preferences: theme, **time zone**, and turning on the **application picker** and **update set picker**
User menu	**Impersonate user**, Elevate role, Log out

Application Navigator

- **Applications** are top-level menus; **modules** are the links inside them.
- **Favorites** (star): create them by dragging a record or list into the navigator, or from a form's additional actions menu. Edit a favorite's label with the **pencil** icon.
- **History** (clock): return to something you viewed recently.
- Type in the filter box to narrow the navigator. `table_name.list` opens a list; `table_name.form` or `table_name.do` opens a new form; typing it in capitals opens a new tab.

Form header icons

- **Additional actions** (three-bar icon, sometimes called the hamburger): the form context menu. You can also right-click the header.
- **More options (…)**: tags, and toggling the template bar.
- Paperclip: attachments.

Lists and filters

A **list** shows many records; each row is a record and each **column is a field**. A **form** shows one record.

> - A filter condition has three parts: **field**, **operator**, and **value**. The operator is always a choice list.
> - The fields offered depend on the table, the columns on it, and the user's access rights.
> - Build two conditions that must both be true with **AND**, then click **Run**. OR means either can match.

Quick filter technique	How
Show only one value	Right-click a value in the list > **Show Matching** (or Filter Out to exclude it)
Column search	Click the list magnifier, type under a column. `*email` means **contains**, `email%` means starts with, and `%email` means ends with. Text with no wildcard runs a greater-than-or-equal search.
Breadcrumbs	Click a breadcrumb to remove the conditions after it
Natural Language Query	Type a plain-language phrase to filter the list

Context menus and tags

Open a context menu by **right-clicking** a list, form header, or column heading, or by clicking the **context menu icon**. Tags can be added from More options (…) > Add Tag. Tag visibility can be **Everyone**, **Groups and users**, or **Me**.

2. Instance configuration 11%

Branding, properties, and plugins

System Properties > Basic Configuration UI16 controls the instance look, including: banner image, **browser tab title**, **base theme**, **header background color**, banner text color, **module text color**, and background and divider colors. System Properties > My Company holds the company **banner image** and **banner text**.

> **Watch for:** default calendar, font style, startup sound, animation style, and preferred browser are not Basic Configuration UI16 settings.

Users fix their own date and time display by setting their **time zone in user preferences** (gear icon). Don't change the instance time zone for one person.

Plugins

Plugins add applications or functionality. Activate them from System Definition > Plugins (or All Available Applications).

- You can't activate a plugin directly in production. Test it on a non-production or **Personal Developer Instance**, then request it for production through Now Support.
- When a plugin depends on others, the system tells you before activation and activates the dependencies.
- Once activated, a plugin **can't be deactivated**. Some offer demo data at activation.

Admin roles, high security, and impersonation

Role	Grants
`admin` (System Administrator)	Access to all system features, functions, and data, because admins can override ACL rules and pass all role checks. For sensitive data such as HR records, ServiceNow recommends custom admin roles instead.
`security_admin`	Elevated role needed to change high security settings and ACLs. Activate it with **Elevate Role** from the user menu.
`impersonator`	Lets a non-admin impersonate other users, but not admin users

Elevation lasts for the session. To return to normal admin security, **log out and back in**.

Common security modules: System Properties > Security, System Security > High Security Settings, and System Security > Access Control (ACL).

Impersonation

- Start it from the **user menu in the banner**. Use it to test what a user sees.
- Return to your own account with **End Impersonation**.
- Impersonation is logged when the `glide.sys.log_impersonation` system property is true.

Forms, fields, and UI Policies

Tool	Open it from	Use it for
Form Layout	Context menu > Configure > Form Layout	Slushbucket: add, remove, and reorder fields and sections
Form Design	Context menu > Configure > Form Design	Drag-and-drop designer; can also create new fields

Both can add fields and change field order. The Form Designer has three main parts: the **page header**, the **field navigator**, and the **form layout**.

- **Sections** display as tabs (when the user has tabbed forms on) once you **give the section a name**.
- To change a field's label, right-click the label and choose **Configure Label**, or edit it from **Configure Dictionary**.
- Templates are opened from the form header's More options (…) icon.

Field types

Need	Type
Free text with letters, numbers, special characters	String
Drop-down of predefined options	Choice
Look up a record in another table	Reference
Checkbox	True/False
Date and time	Date/Time

UI Policies

A UI Policy is the **no-code** way to make fields **mandatory**, **read-only**, or **hidden** based on conditions. Example: make Resolution code mandatory when State is Resolved, and hide a

checkbox unless the user has a role. UI Policies run on the client, after Client Scripts, and only for form interactions.

3. Configuring applications for collaboration

20%

Users, groups, and roles

Object	Definition	Table
User	A person who can log in	`sys_user`
Group	A set of users who share a common purpose (a team)	`sys_user_group`
Role	A collection of permissions	`sys_user_role`

> - Best practice: assign roles to **groups**, then add users to groups.
> - Roles can contain other roles.
> - Setting up a new IT worker: **create the user**, then **add them to each group** whose work they'll do.

If an IT user only sees self-service, they're missing the `itil` role, usually because they aren't in a group that grants it. A manager sees nothing in **My Groups Work** if they aren't a *member* of those groups; being listed as the group's manager doesn't count.

Tasks, assignment, and the activity stream

Task work can be assigned to **users** and **groups**. **Assignment rules** automatically decide which group gets which kind of task.

Benefits of assigning to a group: members can pick work by SLA urgency or highest priority, and the group can route work by on-call schedule, skills, and availability.

- The **Activity** section (activity stream) shows the most recent updates and the history of work on a record. Use its **funnel icon** to show only work notes.
- On a list, the activity stream icon opens a frame with live updates for records you're watching.
- **My Approvals**: right-click an approval in the list and choose Approve or Reject.

UI Actions

UI Actions add **buttons, links, and context menu items** to forms and lists.

Form UI Actions	List UI Actions
Form buttons, form context menu items, form links	List buttons, list context menu items, list choices, list links

Notifications

Notifications inform users about activity. They're created in **System Notification > Email > Notifications** and can be triggered by record inserts and updates or by **events**.

Tab	Defines
When to send	Record inserted or updated, or event fired; conditions
Who will receive	Users, groups, and **users/groups in fields**. Dot-walk from Caller to Caller.Manager to include the caller's manager.
What it will contain	Subject, message, and an **email template** for reusable content like a company banner

Reporting and Performance Analytics

Create reports from **Reports > View / Run**, from **Reports > Create New**, or by right-clicking a list column header.

Report designer section	Purpose
Data	Source table or report source, and filters
Type	Visualization: list, bar, horizontal bar, pie, donut, semi-donut, speedometer, line, and more (25+)
Configure	**Grouping and calculations**
Style	**Colors, titles, legend layout**

- Click **Run** to see changes without saving.
- Best practice: copy a report before editing it, using the dropdown next to Save (**Insert and Stay** / Save as).

- Reports on Service Catalog fulfillment of a specific item use **Requested Item [sc_req_item]**.
- Pull in data from a related table with **dot-walking**. If the same complex data set is reused often, give users a **report source**. If data lives in two tables and dot-walking can't reach it, create a **database view**.
- Share reports with yourself, groups, or everyone; schedule them to be emailed; add them to dashboards.

Performance Analytics

Use Performance Analytics when you need data **trended over time** and **forecast** into the future, using scheduled snapshots, KPIs, and dashboards. Standard reports show the data as it is right now. Learn it through a Performance Analytics course on Now Learning.

4. Self-service and automation 20%

Service Catalog

Build	When
Catalog item	One orderable good or service. Options such as color or RAM are **variables**.
Order guide	Several related items ordered together in one request (new employee: laptop, monitor, phone, business cards)
Record producer	A catalog form that creates a record in another table, like an **incident** from a "Get Help" item. Skips the request and requested item.
Variable set	A reusable group of variables added to many items (four shared mandatory fields across 80 items)

Use **Catalog Builder** to create catalog items and record producers through a guided interface.

> - Request flow: **Request (REQ, sc_request) → Requested Item (RITM, sc_req_item) → Catalog Task (SCTASK, sc_task)**.
> - One request has many requested items; one requested item has many approvals and many catalog tasks.
> - Control who can see an item with **user criteria** (Available for / Not available for).
> - One item needed in two catalogs? Create it once and **publish it to both catalogs**.

A manager approval is skipped when the requester's user record has **no manager**. When testing an approval flow: make sure the latest flow is active, impersonate the requester, and confirm the requester has a manager. On instances using legacy workflows, the approval threshold lives in the **Service Catalog Request** workflow.

Flow Designer

Flows automate business logic such as approvals, tasks, notifications, and record operations with little or no code. Open it from **Process Automation > Flow Designer**.

- A flow starts with a **trigger** (record, schedule, or application) and a **condition**.
- **Actions** do the work. Common core actions: Create Record, Look Up Record, Ask for Approval, Wait for Condition.
- Each action's output becomes a **data pill**. Drag data pills into later steps to reference record data. If a later action changes a data pill's value, the new value is used for the rest of the flow.
- Flows handle multi-level approvals with notifications at each level.

Advantages: less manual scripting, reduced technical debt, and smooth third-party integration through **IntegrationHub spokes**. Spokes are available from the **ServiceNow Store**. **Process Automation Designer** combines cross-enterprise workflows into one unified process.

Knowledge Management

Knowledge Management lets the organization create, categorize, review, approve, and browse information in one shared place.

- Access is controlled with **user criteria** on the knowledge base's **Can Read** and **Can Contribute** lists. Articles can also have their own Can Read user criteria to narrow access further.
- If a knowledge base has no user criteria, **all users can read it** by default. Newer instances can block this with a security property, so always set Can Read.
- With **Match All** selected, a user must meet every condition (company AND department AND group).
- Importing articles requires **Can Contribute** access.
- Check a specific user's access with **User Criteria Diagnostics**.
- **Workflows** control publishing and retiring articles. Article states: Draft, Review, Published, Retired.
- `knowledge_admin` manages multiple knowledge bases; a knowledge manager owns a specific one.
- Continuous improvement: flag articles, mark helpful, rate with stars, and comment.

Virtual Agent delivers help, knowledge, and common tasks through a conversational messaging interface.

Scripting basics

A **Business Rule** runs on the server when a record is displayed, inserted, updated, deleted, or when a table is queried. Its **When** options are **before**, **after**, **async**, and **display**. A **Client Script** runs in the browser.

5. Database management 27%

The biggest domain. Spend real time here, and build tables, relationships, and ACLs in your PDI.

Tables, fields, and relationships

A table holds records (rows); each **field** (column) holds one piece of data for a record. The primary operations on tables are **create, read, write, and delete**.

Term	Meaning
Core table	Ships with the platform (User, Task, Incident)
Custom table	Created by you (`u_` prefix in global; scope prefix such as `x_` or `sn_hr_` in a scoped app)
Base table	Not extended from any other table (Task)
Parent table	Has tables extended from it; may itself extend another table
Child table	Extends another table and inherits its fields

A table can't be both core and custom. Any other combination is possible.

Worked example: Incident extends Task, and Super Incident extends Incident. Task is a **base** and **parent** table. Incident is both a **child** and a **parent**. Super Incident is a **child**.

For a new app whose tickets get assigned to a team, **extend Task** and add your own fields, so you inherit assignment, state, and work notes.

Relationships

Relationship	Example
One-to-many	One request has many requested items (a reference field)
Many-to-many	Vendors sell many products; products come from many vendors. Related lists show on both tables.
Extension	Incident extends Task
Database view	Joins two tables for reporting

- Create tables with **System Definition > Tables** or **Tables & Columns**.
- See a table's relationships with the **schema map**: System Definition > Tables, open the table, Related Links > Show Schema Map.
- The **System Dictionary** defines every table and field. Dictionary overrides change a field's properties on a child table.
- A new custom table gets four ACLs automatically: **create, read, write, delete**.

Access Controls (ACLs)

ACLs control access to records and fields for each operation: create, read, write, delete. Permissions are set with **roles**, **conditions**, and a **script** that sets `answer` to true or false.

Name	Secures
`incident.None`	All records on the table (row level)
`incident.*`	All fields on the table
`incident.short_description`	One field

Let itil users create incidents: Name `incident.None`, Operation **create**, Role **itil**.

> - Table (row) ACLs are evaluated before field ACLs, so a user must have access to the table before field access is considered.
> - If a row rule and a field rule both apply, **both must pass**.
> - Table ACLs go from specific to general: the table, then its parent table, then the wildcard (`*`).

A user who can see a module but gets a blank page on Create New likely has read but not create access on that table.

Data Policies

Data Policies enforce mandatory and read-only rules on the **server**, for every way data arrives: forms, import sets, and web services. Use them for rules like "Short description is mandatory on insert everywhere" or "Number is read-only for all users." A Data Policy can also be applied as a UI Policy on forms.

CMDB and CSDM

Table	Role
`cmdb`	Base Configuration Item table, top of the CMDB hierarchy
`cmdb_ci`	Configuration Item table; CI classes extend it (cmdb_ci_server, cmdb_ci_computer)
`cmdb_rel_ci`	Relationships between CIs

- Defining **suggested relationships** between CI classes makes relating CIs easier, because the platform suggests which types fit.
- The **CI dependency view** graphically shows a CI's infrastructure relationships. Right-click a relationship line and choose **View Relationship Form**; right-click a CI for options like View Form and View Related Tasks. Service desk agents use it to trace from a service to the CIs with active issues.
- On a CI form, **Related Items** shows the other CIs it's related to.
- Populate the CMDB with Discovery, Service Mapping, import sets, integrations, or manual entry.

The **Common Service Data Model (CSDM)** is the standard, shared set of service definitions across ServiceNow products. It's the blueprint for mapping your IT services into ServiceNow and supports service-level reporting.

6. Data migration and integration 15%

Import Sets

Import data with the **System Import Sets** application. The steps are: **load data**, **create a transform map**, **transform**, and **clean up** the import set table.

1. **Load Data** (first step) brings the source into an **import set row table**, the staging area.
2. A **transform map** defines where incoming columns go on the target table (field mapping).
3. **Transform** copies data from staging to the target.

Mapping tool	When
Automatic mapping (Auto Map Matching Fields)	Source and target field names match
Mapping Assist	Drag-and-drop linking of source to target fields

> **Coalesce** fields are the unique key. Match found: the record is updated. No match: a new record is inserted. **No coalesce field**: every row is inserted as new, and nothing is updated.

Data source types: local files (CSV, Excel, XML), files on a network server (FTP, HTTP), **JDBC** database connections, and **LDAP**. Target tables can be in the current scope, in global, or in other scopes that allow write access.

Best practice: plan time before importing to clean out obsolete or inaccurate data.

Update Sets

Update sets capture configuration changes so you can move them between instances.

1. Create a named update set and make it current.
2. Make your changes, then mark the set **Complete**.
3. On the target: retrieve, **preview**, then **commit**.

> - Never use the **Default** update set to move changes between instances.
> - Update sets carry configuration, not task or process data. Not captured by default: **data records** and **homepages and content pages**. Add a homepage manually (Homepage Admin > Pages, right-click > Unload Portal Page) or a dashboard with **Unload Dashboard**.
> - Keep update sets small to reduce conflicts and make changes easier to review.
> - Moving several at once? **Batch** them.
> - Move data with import sets or XML export and import, not update sets.

Table names to memorize

The exam asks for exact names. Type each one followed by `.list` in your PDI.

Label	Table
User	`sys_user`
Group	`sys_user_group`
Role	`sys_user_role`
Task	`task`
Request / Requested Item / Catalog Task	`sc_request` / `sc_req_item` / `sc_task`
Catalog Item	`sc_cat_item`
Base CI / CI / CI relationship	`cmdb` / `cmdb_ci` / `cmdb_rel_ci`
System Properties / Business Rules / Client Scripts	`sys_properties` / `sys_script` / `sys_script_client`
Dictionary	`sys_dictionary`

Rapid-fire recall

Say the answer out loud, then open it to check.

- Three parts of a filter condition?

  Field, operator, value.

- Column search for "contains"?

  Start the term with `*`.

- Where do you turn on the update set picker?

  The gear icon (user preferences).

- Role needed for high security settings, and how do you drop it?

  security_admin, elevated for the session. Log out and back in.

- How do you return from impersonation?

  End Impersonation from the user menu.

- Group vs. role?

  A group is a set of users; a role is a collection of permissions. Assign roles to groups.

- Report designer section for grouping? For colors and titles?

  Configure. Style.

- Trend and forecast data over time?

  Performance Analytics.

- Laptop, phone, and badge in one request?

  Order guide.

- Catalog form that creates an incident?

  Record producer.

- Same four variables on 80 items?

  One variable set added to every item.

- Why would a manager approval be skipped for one requester?

  The requester's user record has no manager.

- Where does action output live in a flow?

  In data pills.

- Knowledge base with no user criteria: who can read?

  All users, by default.

- Tool to check one user's knowledge access?

  User Criteria Diagnostics.

- Business Rule When options?

  Before, after, async, display.

- ACL to let itil users create incidents?

`incident.None`, operation create, role itil.

- No-code mandatory field on forms only? On all data entry?

  UI Policy. Data Policy.

- Three key CMDB tables?

  cmdb, cmdb_ci, cmdb_rel_ci.

- Blueprint for mapping IT services into ServiceNow?

  CSDM.

- Import steps in order?

  Load data, create transform map, transform, clean up.

- What happens with no coalesce field?

  Every row is inserted as a new record.

- Two things not in an update set by default?

  Data records and homepages.

- Moving several update sets at once?

  Batch them.
